

**D 133521**

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Name.....

Reg. No.....

**FIRST SEMESTER (CUFYUGP) DEGREE EXAMINATION  
NOVEMBER 2025**

Applied Physics/Physics

APH1FM105/PHY1FM105—PHYSICS IN DAILY LIFE

(2024 Admission onwards)

Time : One Hour and a Half

Maximum : 50 Marks

**Section A***Answer all questions.**Each question carries 2 marks.**Ceiling 16 marks.*

1. What is the principle behind the heating process of a microwave oven ?
2. Explain how the fresh air fan functions in a kitchen.
3. What role does copper play in kitchen utensils ?
4. Describe how noise is produced by a dishwasher.
5. What is the principle behind a weighing scale ?
6. Explain why metal objects can feel colder than plastic ones.
7. What are the physics behind spin bowling in cricket ?
8. Why is airflow separation important in football physics ?
9. Explain the importance of boundary layer effects in football.
10. How does the hot spot technology work in cricket ?

**Section B***Answer all questions.**Each question carries 6 marks.**Ceiling 24 marks.*

11. Discuss how an induction cooktop utilizes electromagnetic fields to heat cookware.
12. Explain the physics behind the goalkeeper's throw in football.

**Turn over**

13. Compare the working principles of Hawkeye and Snicko technologies in cricket.
14. Describe how turbulence affects the trajectory of a football during a game.
15. What are the common methods to reduce energy waste in the kitchen ?

### Section C

*Answer any **one** question.  
The question carries 10 marks.*

16. Discuss in detail the physics of pace bowling in cricket, including seam, air resistance, and swing.
17. Explain the principles of a banana kick in football, including the role of the Bernoulli effect and Magnus effect.

(1 × 10 = 10 marks)